

Rizelle B. Bautista

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[AWS Developer Associate](#) • [AWS AI Practitioner](#) • [AWS Cloud Practitioner](#) • [AWS Serverless & Incident Response](#)

PROFESSIONAL SUMMARY

AI Software Engineer with proven production experience building end-to-end LLM-powered solutions — from problem framing and data readiness through evaluation, rollout, and iteration. At WebPuppies (Singapore), independently architected an enterprise agentic AI platform using Amazon Bedrock, multi-step orchestration, and serverless AWS infrastructure now live in production and central to the company's AWS APN partnership. Built and shipped hallucination-reduction frameworks, iterative refinement loops, and reusable LLMOps patterns. Experienced partnering directly with business stakeholders to translate ambiguous requirements into scoped, measurable AI solutions. Strong production-readiness instincts: observability, monitoring, quality gates, CI/CD, and safe rollouts. Familiar with security, PII handling, and responsible AI controls for enterprise environments.

PROFESSIONAL EXPERIENCE

Software Engineer — AI & Cloud | *WebPuppies, Singapore (Remote)*

Dec 2025 – Present

Pioneer Lead — LLM Application Architecture, Agentic Systems & Full-Stack AI Deployment

- Led solution architecture and end-to-end development of a production enterprise LLM application from scratch — translating business requirements into a scalable agentic system using Amazon Bedrock (Claude), AWS Lambda, Step Functions, API Gateway, S3, and DynamoDB — following full SDLC from requirements through deployment, now integrated into the company website.
- Engineered a multi-step LLM evaluation and hallucination-reduction framework — structured JSON output schemas enforce response format, rule-based validation checks every AI-generated field against business logic, and failed outputs trigger automatic re-prompting with failure context injected — achieving near-zero error rate on enterprise financial outputs delivered to business stakeholders.
- Built full-stack AI interfaces: React/Next.js frontend integrated seamlessly with serverless AI backend APIs via AWS API Gateway — exposing agentic capabilities to end users with high availability, low latency, and production-grade reliability.
- Defined and implemented LLMOps best practices: end-to-end CI/CD pipelines via GitHub Actions, AWS CodeBuild, and CodePipeline; IaC-style AWS SAM templates for environment parity across dev, staging, and production; automated testing gates before every deployment — zero-downtime releases across all environments.
- Applied AWS Bedrock Guardrails, IAM least-privilege policies, and Secrets Manager across all LLM outputs and infrastructure resources — enforcing responsible AI, compliance, and data security controls in a production environment, directly analogous to banking security standards.
- Designed event-driven agentic orchestration on AWS Step Functions capable of processing 120,000+ row datasets per run with real-time status updates, automated error handling, and retry logic — demonstrating production multi-step agent reliability at enterprise scale.
- Collaborated directly with the Head of Technology throughout the full SDLC — requirements gathering, architecture review, iterative development, and deployment sign-off — platform became the key technical deliverable for the AWS APN Standard Tier partnership application.

President | *AWS Learning Club — JRU Chapter*

Mar 2025 – Present

- Defined engineering best practices and cascaded AI design patterns to 150+ members — conducting structured technical sessions on cloud computing — guiding 15+ to AWS certification, the highest certified count across all PH AWS Cloud Club chapters.
- Co-organized AWS Student Community Day Mega Manila 2025 drawing 400+ participants, coordinating cross-team Agile delivery via Jira with sprint planning and milestone tracking.
- Established training partnerships with DataCamp, Tutorialz Dojo, and KadaKareer and led hands-on AWS Academy workshops — building a structured learning pipeline that equipped 200+ members with cloud and AI skills, resulting in the highest AWS-certified membership count across all PH AWS Cloud Club chapters.

Web Developer & Data Analyst Intern | *Analytics & AI Association of the Philippines*

Oct 2024 – Present

- Built and shipped two production full-stack websites (aap.ph, aicon.ph) using lowcode tools (WIX, WildApricot CRM) with third-party API integration (Helixpay), Google SEO, data analysis, and stakeholder-facing strategic reporting — independent end-to-end ownership from development through production deployment. Created data analysis report and visualization using Python, Power BI, and LookUp.
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KEY PROJECTS — LLM APPLICATIONS & AI ENGINEERING

Lily — Multi-Agent AI System with RAG, Evaluation & Hallucination Controls | 2024–2025
Undergraduate Thesis | Project Manager, Full-Stack Developer & AI Engineer

- Architected and built a production multi-agent AI application integrating five simultaneous AI pipelines: Faster-Whisper STT with DSP noise reduction, CNN emotion detection (85% accuracy), Hybrid Adaptive RAG (FAISS vector search + pgvector), dual-mode LLM orchestration (Groq Llama 3.1-8B online / Ollama Llama 3.2:1b offline), and dual TTS — a complex agentic system directly analogous to enterprise LLM application architecture.
- Engineered a production LLM evaluation framework — every RAG response was validated by a secondary Groq model acting as an AI content validator, checking factual consistency against source documents before delivery — reducing hallucinations to near-zero in a real student-facing production environment.
- Designed advanced RAG retrieval with Hybrid Adaptive strategy: online mode uses cloud embeddings + pgvector for scalable retrieval; offline mode uses local FAISS for sub-millisecond lookup without network dependency — with automatic failover between modes achieving 100% service continuity.
- Built full-stack application: Python/FastAPI backend exposing AI capabilities via RESTful APIs, React admin dashboard for knowledge base management and system monitoring, and Supabase PostgreSQL for structured data — deployed to AWS with full production availability.
- Maintained 2–4 second end-to-end LLM response times under continuous production load for 1,000+ projected users by orchestrating five AI pipelines on constrained edge hardware with resource prioritization — demonstrating production AI performance engineering under real constraints.
- Reduced solution cost 80% (100,000 PHP to 30,000 PHP) versus industry-standard solutions through hybrid cloud-edge architecture and FinOps-aligned AI service selection.

LangChain Agentic Chatbot — Full SDLC with Persistent Memory | *Python, LangChain, LangGraph, React — LangChain Academy Capstone* 2024

- Built a production multi-turn agentic AI application using LangChain orchestration, LangGraph for stateful agent graph management, and persistent cross-session memory — implementing the full SDLC from requirements through deployment, with prompt templating, output parsers, memory buffers, and tool-use patterns using LangChain Expression Language (LCEL).

Amazon Bedrock LLM Evaluation & Prompt Engineering Platform | *Amazon Bedrock, Claude AI, Python, Node.js, AWS Lambda* 2024

- Developed multi-step prompt engineering pipelines and iterative LLM evaluation loops within Amazon Bedrock — each output validated against defined business rules before acceptance, with automatic re-generation on failure — producing deterministic, enterprise-ready financial analysis outputs with measurable accuracy standards.

Highly Available Web Application | *AWS: VPC, EC2, Aurora RDS, ALB, Auto Scaling, CloudWatch, IAM, Secrets Manager* 2024

- Architected a fault-tolerant multi-tier application across multiple AWS Availability Zones with Auto Scaling, ALB, Multi-AZ Aurora RDS, and CloudWatch-based incident response — validated through load testing, demonstrating production infrastructure reliability standards applicable to banking environments.

THRIVE — Inclusive Job Platform with NLP Matching & Accessibility | *React Native, Supabase, PostgreSQL, Python, STT/TTS, NLP | PLDT Season 4 Finalist* | 2024

- Team Leader and Full-Stack Developer for an AI-powered inclusive job platform — integrated NLP-based semantic similarity matching,
- STT/TTS accessibility features validated by PDAO Cainta, and role-based dashboards; conducted iterative user testing with PWD communities across multiple regions.

FRAPS Solar Works — Database Management System | *Java, MySQL, XAMPP, NetBeans* | *Full SDLC — Requirements through Delivery*

- Sole developer responsible for full SDLC: designed and normalized a relational MySQL database schema with multi-role access control; built inventory tracking, service management, and sales reporting modules — eliminating manual data management inefficiencies for a client-serving business.

TECHNICAL SKILLS

LLM Application Development: Amazon Bedrock (Agents, Guardrails), LangChain (Academy certified), LangGraph, multi-agent orchestration, agentic patterns (ReAct, tool-use, MCP concepts), Model Context Protocol awareness, full SDLC for LLM apps

Prompt Engineering & Evaluation: Structured output schemas, iterative refinement loops, chain-of-thought, few-shot prompting, LLM evaluation frameworks, hallucination detection and reduction, AI content validation pipelines, accuracy measurement

RAG & Vector Architecture: Hybrid Adaptive RAG (FAISS + pgvector), document chunking strategies, sentence transformer embeddings, cosine similarity retrieval, knowledge base management; learning Pinecone, Milvus

LLMOps / MLOps: CI/CD for AI (GitHub Actions, AWS CodeBuild, CodePipeline), AWS SAM IaC, automated testing gates, CloudWatch monitoring and alerting, incident response (AWS certified), environment parity across dev/staging/prod

Programming: Python (Expert), SQL (PostgreSQL, MySQL, Aurora RDS, SQL Server, Oracle) — data transformation, analytics, schema design, query optimization; JavaScript/TypeScript, Java, PHP, Bash

Full-Stack Development: React, Next.js, React Native, Streamlit — production UIs integrated with AI backend services; FastAPI, Node.js, RESTful APIs, Microservices

AWS & Cloud: Lambda, Step Functions, API Gateway, S3, DynamoDB, EC2, RDS (Aurora), CloudWatch, IAM, Secrets Manager, Bedrock, Elastic Beanstalk, Auto Scaling, ALB, VPC, EventBridge, SAM CLI, CodeBuild, CodePipeline

Security & Compliance: AWS IAM least-privilege, Secrets Manager, Bedrock Guardrails, VPC security groups, incident response procedures — production security controls applicable to regulated banking environments

AI Coding Assistants: Cursor, Claude Code, GitHub Copilot, Open AI— daily AI-accelerated development workflow

Databases & Vector Stores: PostgreSQL, MySQL, Aurora RDS, DynamoDB, Supabase, Databricks, FAISS, pgvector, SQL Server, Oracle, Firebase

Methodologies: Agile/Scrum, SDLC, requirements gathering, unit testing for AI, code reviews, technical documentation, stakeholder presentations

CERTIFICATIONS

- AWS Certified Developer – Associate
- AWS Certified AI Practitioner
- AWS Certified Cloud Practitioner
- AWS Microcredentials: Serverless Demonstrated & Incident Response
- LangChain Academy — LangChain & LangGraph Curriculum
- CompTIA ITF+
- Currently pursuing: AWS Certified Solutions Architect – Associate

AWARDS & RECOGNITION

- Hack-Nation AI Global Hackathon 2026 — Selected alongside participants from MIT, Harvard, and Stanford University
- 2nd Place — Globe Telecom Code-To-Career Hackathon 2025 | Python data engineering for real-world solution development
- Finalist — PLDT & Smart Innovation Generation Season 4 | Led full-stack development of THRIVE as project manager
- Super AI 2025 Singapore Delegate — One of 100 students worldwide selected to attend the largest international AI summit

EDUCATION

Bachelor of Science in Computer Engineering | *Jose Rizal University, Philippines* 2022 – 2026
Dean's Lister